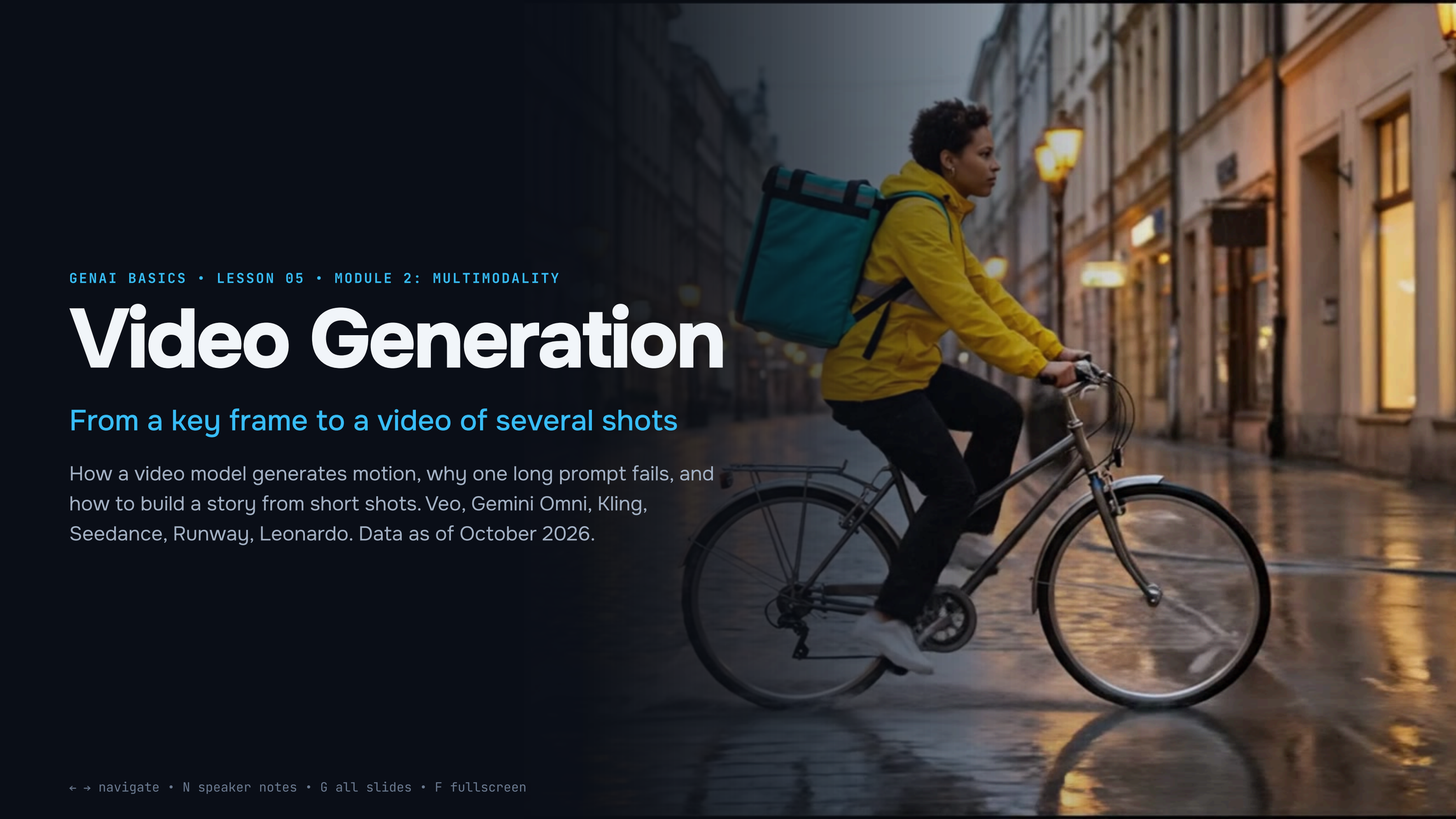
GENAI BASICS • LESSON 05 • MODULE 2: MULTIMODALITY

Video Generation

From a key frame to a video of several shots

How a video model generates motion, why one long prompt fails, and how to build a story from short shots. Veo, Gemini Omni, Kling, Seedance, Runway, Leonardo. Data as of October 2026.



EXAMPLE

A 23-second video built from three 8-second shots



3 shots

each generated from an **approved key frame** (image-to-video)

8+8+8 → 23 s

joined with two 0.5 s transitions

≈ $10

3 × 8 s × $0.40, **Veo 3.1** list price, 720p, failed attempts not included

> Next: where to generate, how the model works, why one prompt for the whole story fails, and a 6-step pipeline.

WHERE TO GENERATE

Three access layers for video models: chat, web studio, API

Chat apps

video inside a chat

Gemini app · Google Flow · Grok Imagine

CONTROL	prompt text only; Flow adds first/last frame and extension
BILLING	subscription
USE FOR	ideas, quick drafts
START WITH	**Gemini app**

Web studios

dedicated video UI

Kling · Leonardo · Dreamina / CapCut (Seedance) · Runway · Luma · Hailuo

CONTROL	camera settings, first/last frame, references, several shots per generation
BILLING	credits; small free allowance
USE FOR	individual videos, homework
START WITH	**Kling**

API

code and automation

Gemini API / Vertex AI · Kling API · Runway API · xAI API

CONTROL	everything as request parameters; batch jobs
BILLING	per second of video
USE FOR	pipelines, products
START WITH	**Gemini API**

> One model is often available on all three layers: Veo is in Flow, Leonardo, Runway and the Gemini API (the Gemini app now runs Gemini Omni).
> **ChatGPT does not generate video**: Sora has been shut down.

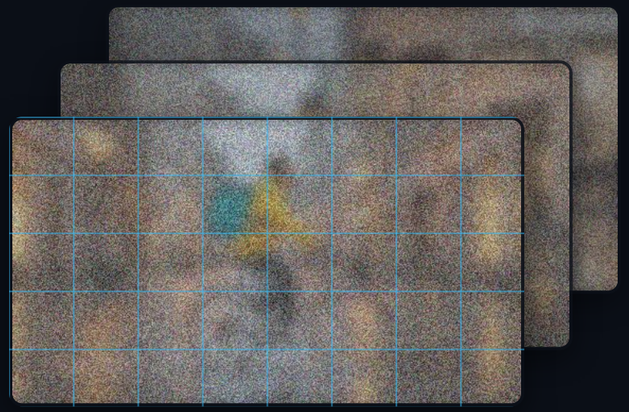
HOW IT WORKS

How a diffusion model generates video frames



1. Noise

Start: random noise for **all frames** of the clip at once



2. Denoising, step by step

The video is cut into pieces “place in frame × moment in time”. Dozens of steps remove noise; the text and the first frame guide every step



3. Frames

24 frames per second that are consistent with each other, so motion looks smooth

There is no physics engine inside. The model has seen millions of videos and knows what motion *usually looks like*. So it fails where exact rules matter: **hands and objects in contact, text, long scenes**.

Schematic: frames from our shot 2, noise added for illustration.

ANTI-PATTERN

Text-to-video, the whole story in one prompt: 4 failure types

TEXT-TO-VIDEO Veo 3.1 • 8 s • text only



PROMPT (SHORTENED)

```
Our friendly Glovo delivery courier enters a cozy
Scandinavian cafe, … picks up a brown paper food delivery
bag, puts it securely inside her turquoise backpack, then
cuts to her riding a city bicycle through the rainy avenue
at dusk … and then camera zooms in to a smartphone map …
```

> Root cause: **3 scenes and 5 actions** requested in 8 seconds.



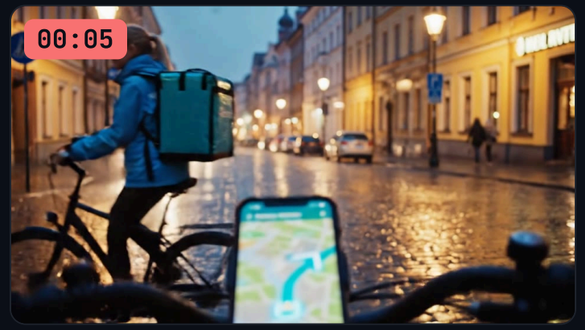
Action skipped

the bag stays on the counter and never goes into the backpack



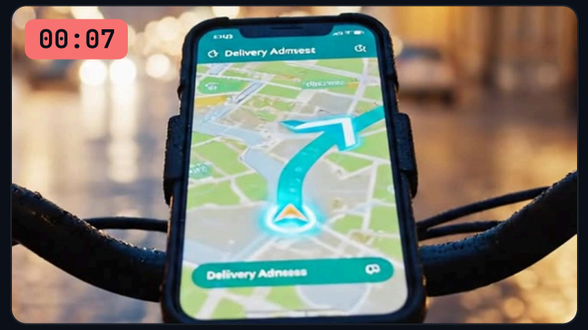
Details drift after the cut

the logo on the box changes: “Glono” → “Glovo”



Scene geometry breaks

handlebars with a phone slide in while the courier is still riding ahead



Garbled text

“Delivery Admest” instead of “Delivery Address”

TWO GENERATION MODES

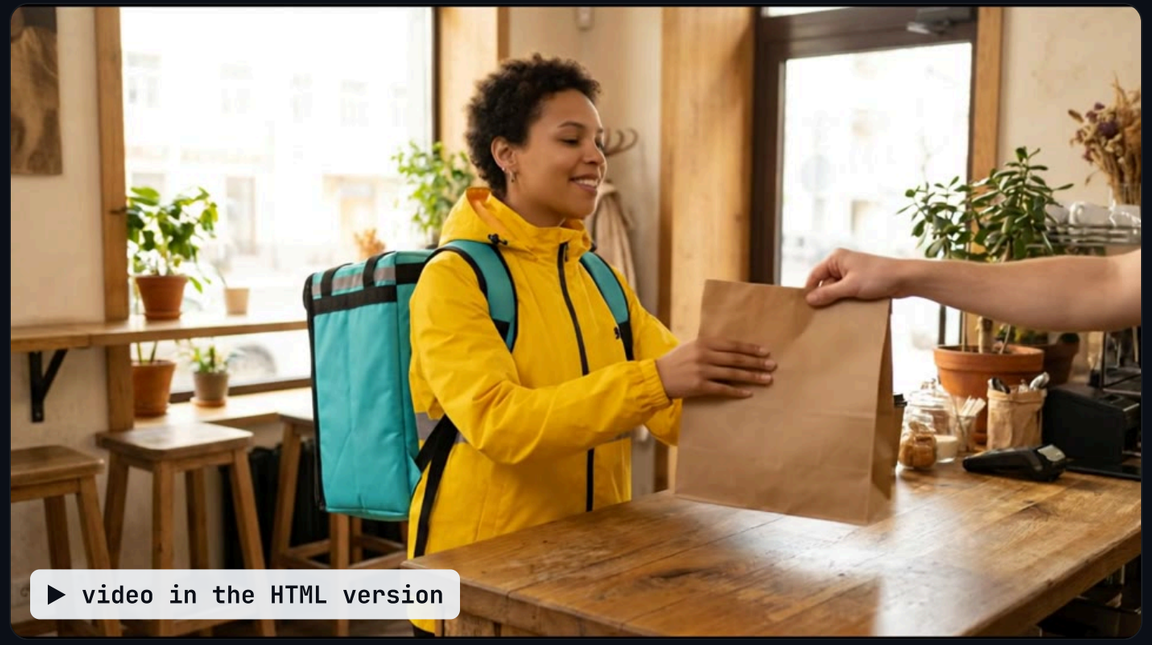
Text-to-video vs image-to-video

TEXT-TO-VIDEO input: text only



Text → The model decides **what is in frame** and **how it moves** → ✗ **two tasks at once**

IMAGE-TO-VIDEO input: key frame + text



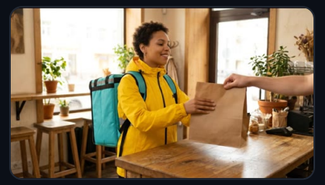
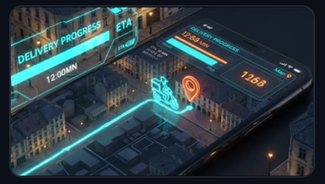
Frame ✓ + text about motion → The model decides only **how it moves** → ✓ **one task**

Rule: fix appearance, light and composition in an image (the L04 skill). The video model handles only the motion.

STEP 1 • STORYBOARD

Storyboard: the whole story before the first generation

The story is split into shots of 4–8 seconds. Each row = one key frame + one prompt + one generation.

#	PREVIEW	TIME	ACTION	CAMERA	KEY FRAME FILE
1		0–8 s	Courier takes the bag from the barista and turns to the exit	medium shot, slow dolly in	`shot1_cafe_pickup.png`
2		7.5–15.5 s	Rides a bicycle down a wet street at dusk	tracking alongside, low angle	`shot2_bike_rain.png`
3		15–23 s	The map route animates towards the destination pin	slow digital zoom in, slight tilt	`shot3_map_route.png`

FULL TEMPLATE (CREATOR TOOLS) **Shot / timing** **Visual** **Voice-over** **Image prompt** **Video prompt** **SFX (sound effects)** **Music**

> **Why short shots:** a long or extended clip accumulates drift, and any defect means regenerating all of it. A short shot is more stable, only that shot is redone, and the budget stays predictable.

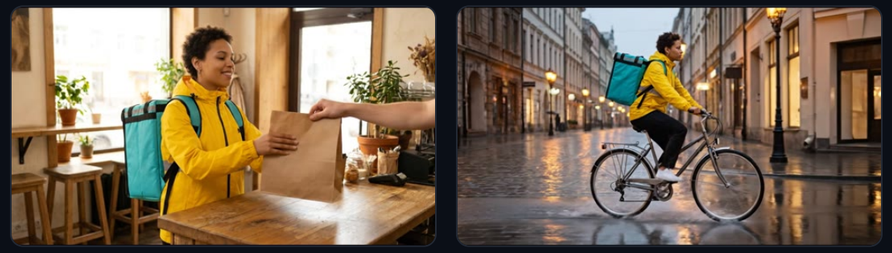
Times include the 0.5 s transition overlaps. 1 minute ≈ 8 shots of 8 s; 1 hour ≈ 450 shots. Keep the storyboard in a spreadsheet (Google Sheets, Excel): it is the work plan, the budget and the acceptance list.

STEP 2 • REFERENCES

Character sheet as the reference for every shot



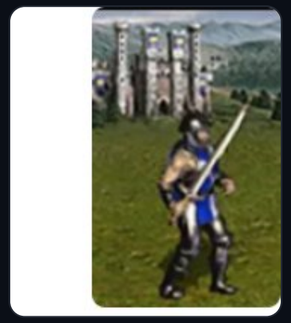
- A reference **narrows the model's choices** for appearance.
- **Veo 3.1:** up to 3 reference images. **Kling 3.0:** Elements, 2–4 images.
- More stable, **not guaranteed**: review every shot.



Result: **the same courier** in shots 1 and 2.

STEP 3 • KEY FRAME (L04 EXAMPLE)

Finalise the key frame in an image chat before generating video



Sprite: game source art



“Make a realistic 16:9 frame from this sprite”

Draft: wrong helmet and tabard



“Bare hands, tabard in white and blue rectangles, a flat ‘kettle’ helmet”

Approved ✓: first frame for the video

Image edit: seconds, cents

one **image** edit in chat

Video regeneration: minutes, $0.40 per second

one **video** regeneration (Veo 3.1, 720p)

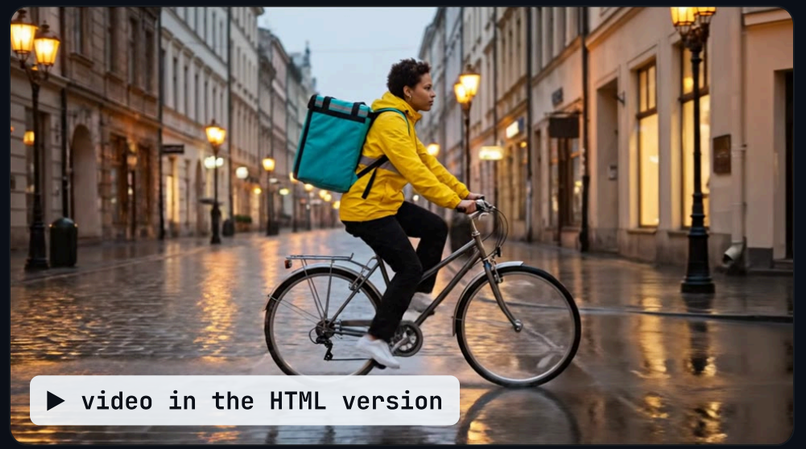
STEP 4 • PROMPT

Video prompt structure: 6 slots

1 Camera · 2 Who · 3 Action · 4 Where & when · 5 Motion details · 6 Light

SHOT 2 PROMPT (VEO 3.1, IMAGE-TO-VIDEO)

1 Dynamic tracking shot moving forward alongside the courier. 2 The same courier with short cropped curly hair 3 pedals her bicycle forward 4 through the rain-slicked city avenue at twilight. 5 Wheels spinning smoothly forward with water spray cutting through shallow puddles. 6 Wet cobblestones reflecting warm shop lights and streetlamps.



Output: shot 2

- **Motion verbs** (pedals, spinning, moving forward): the model needs to know what changes.
- **Do not describe the frame again:** it is already the input. One identifying detail is enough.
- **Describe what should happen.** A negative prompt field (what must not appear) exists only in some tools (Kling, Wan).

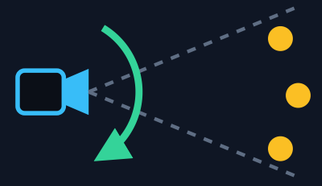
STEP 4 • CAMERA

Camera moves and the prompt terms for them



Fixed camera `Static`

“static camera, locked-off shot”



Rotate in place `Pan`

“slow pan left across the street”



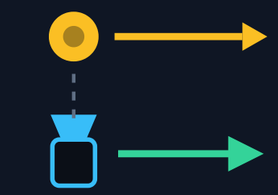
Tilt up / down `Tilt`

“tilt up to the rooftops”



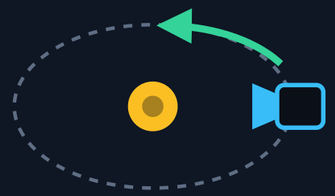
Move closer / away `Dolly in / out`

“slow dolly in to her face”



Move alongside the subject `Tracking`

“tracking shot alongside the courier”



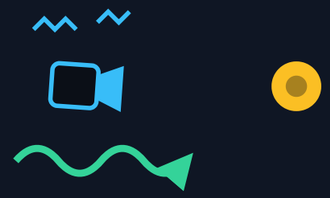
Circle around the subject `Orbit`

“slow orbit around the mug”



Camera rises `Crane`

“crane up over the street”



Handheld, slight shake `Handheld`

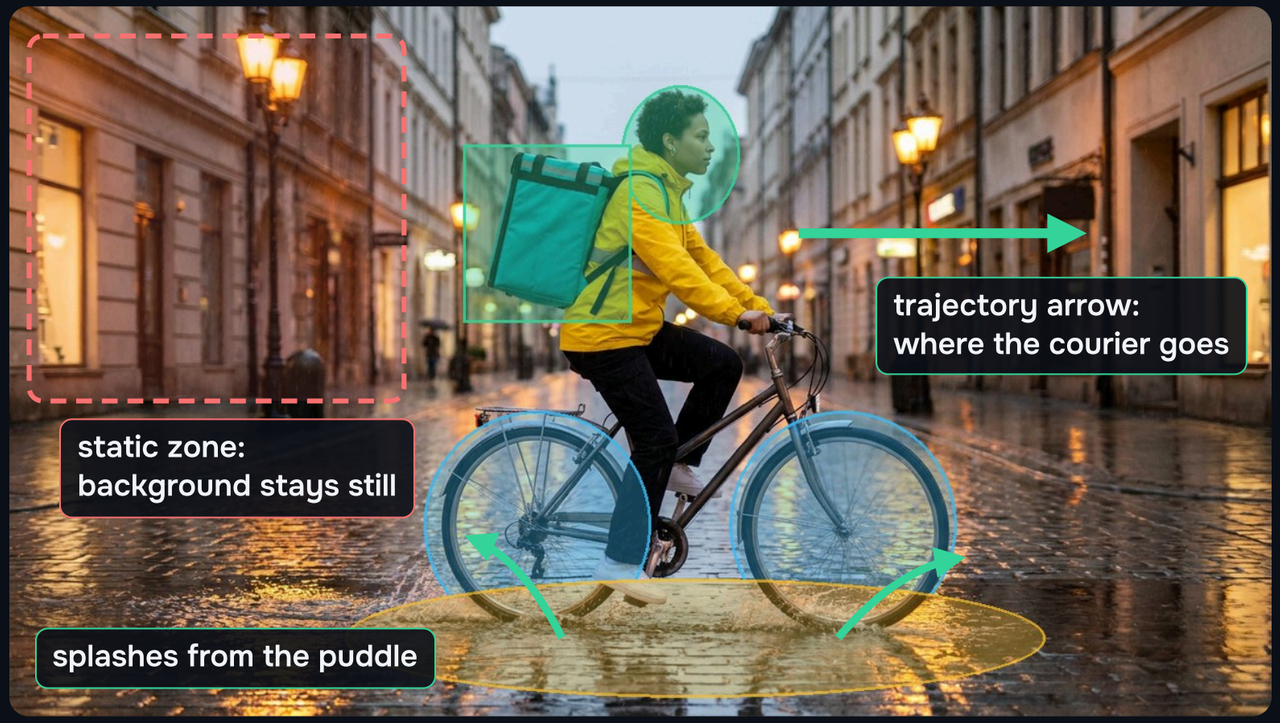
“handheld, slight shake”

> Models were trained on captioned video, so they understand **camera terms**. In chat apps you set the camera with words only; web studios also have dedicated controls (Runway, Kling, Luma).

Diagrams: top view (Pan, Dolly, Tracking, Orbit) and side view (Static, Tilt, Crane). Camera in blue, subject in yellow, motion as a green arrow.

STEP 5 • CONTROL

Controls beyond the prompt: frames, extension, references



Motion Brush (Kling): up to 6 selected elements, each with its own arrow, plus a static zone. Available on Kling 1.x–2.x models; not in the Kling 3.0 feature list.



First + last frame

The model fills in the motion between two images. **Veo 3.1, Gemini Omni, Kling 3.0**



Extend a clip

Veo 3.1: +7 s per step, up to 148 s (720p, not Lite)

Gemini Omni: +10 s per step, up to 40 s



Character references

Veo 3.1 / Fast: up to 3 images, 8 s clip

Kling 3.0 Elements: 2–4 images

STEP 6 • EDITING

Editing: three shots into one 23-second video



8 + 8 + 8 − 0.5 − 0.5 = 23 s

three Veo 3.1 shots and two transitions: a wipe and a fade to black

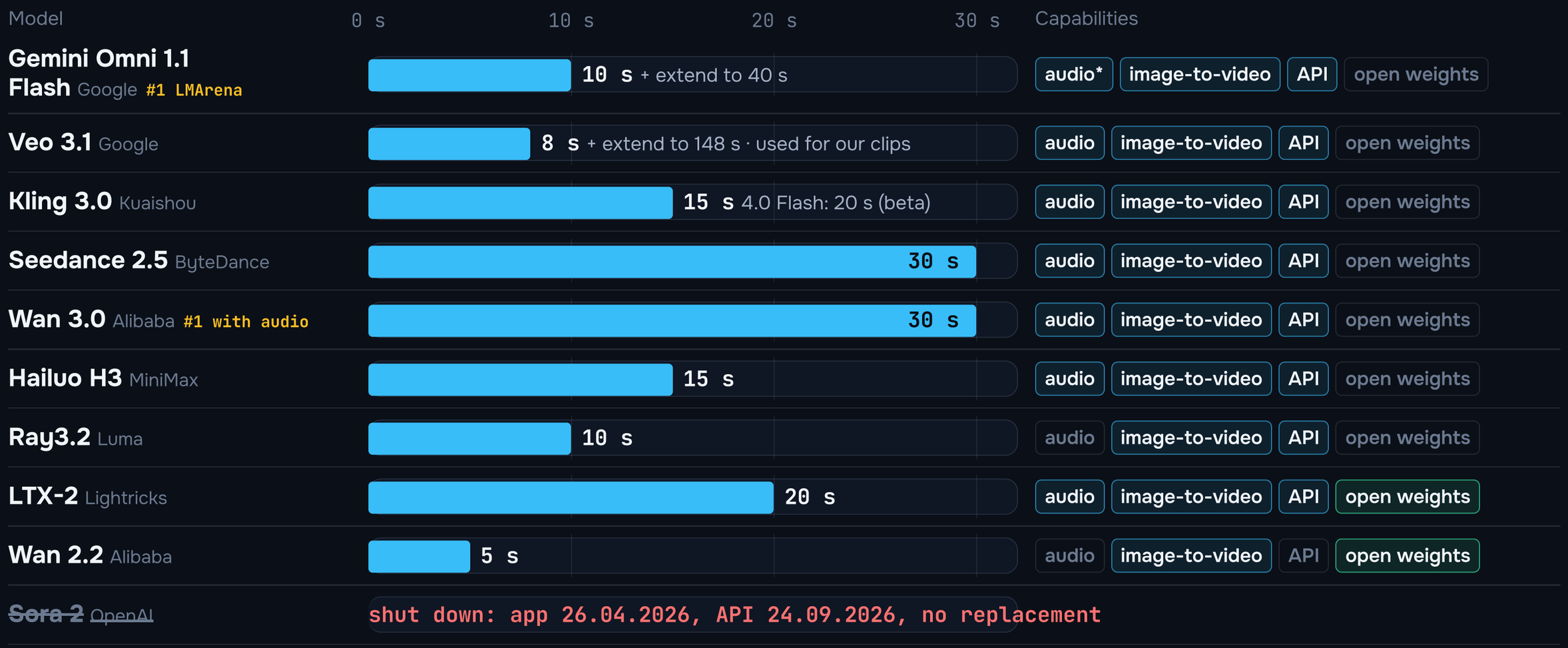
> Join in any editor (CapCut, DaVinci Resolve) or with one command:

```
ffmpeg -i shot1.mp4 -i shot2.mp4 -i shot3.mp4 -filter_complex  "[0]
[1]xfade=transition=wipeleft:duration=0.5:offset=7.5[a];
  [a][2]xfade=transition=fadeblack:duration=0.5:offset=15" final.mp4
```



MARKET • OCTOBER 2026

Video models: maximum clip length and capabilities



Bar: maximum per generation. Highlighted chip: yes; grey: no or not confirmed. * Omni audio per third-party reviews. Rankings (public leaderboards): LMArena (21.09.2026), Artificial Analysis, with-audio category (October 2026).

BREAK

5 minutes for coffee and resting your eyes

05:00

Next: code, cost, acceptance, risks and homework.

REFERENCE • GOOGLE-GENAI

Gemini API: image-to-video in Python

```python
import time
from google import genai
from google.genai import types

client = genai.Client()  # key from GEMINI_API_KEY
op = client.models.generate_videos(
    model="veo-3.1-fast-generate-preview",
    prompt="Tracking shot alongside the courier",
    image=types.Image.from_file(location="shot2.png"),
)
while not op.done:  # wait: ~11 s to ~6 min
    time.sleep(10)
    op = client.operations.get(op)
video = op.response.generated_videos[0].video
client.files.download(file=video)
video.save("shot2.mp4")
```

Long-running operation: **submit → poll status → download**. Generation takes seconds to minutes.

SAME PATTERN IN OTHER APIS

- Kling API
- Runway API
- xAI API (Grok Imagine)

Gemini API, October 2026. No free tier for Veo. Prices are on the next slide.

COST

Generation cost: per clip, per minute, per hour of video

Priced per second of generated video, 720p. One hour of finished video = 3600 s = **450 shots of 8 seconds**.

MODEL	$ PER SECOND	8 S CLIP	1 MINUTE	1 HOUR	1 HOUR, 3 ATTEMPTS PER SHOT
Veo 3.1 Lite Gemini API	$0.05	$0.40	$3.00	$180	$540
Grok Imagine 1.5 xAI API	$0.08	$0.64	$4.80	$288	$864
Veo 3.1 Fast Gemini API	$0.10	$0.80	$6.00	$360	$1,080
Runway Gen-4.5 Runway API	$0.12	$0.96	$7.20	$432	$1,296
Seedance 2.5 Runway API, starting price	$0.20	$1.60	$12.00	$720	$2,160
Veo 3.1 Gemini API	$0.40	$3.20	$24.00	$1,440	$4,320

Our 23 s video: 3 shots × 8 s × 3 attempts = 72 s generated: $28.80 on Veo 3.1, $7.20 on Fast. With 3 attempts per shot you generate three times what you keep.

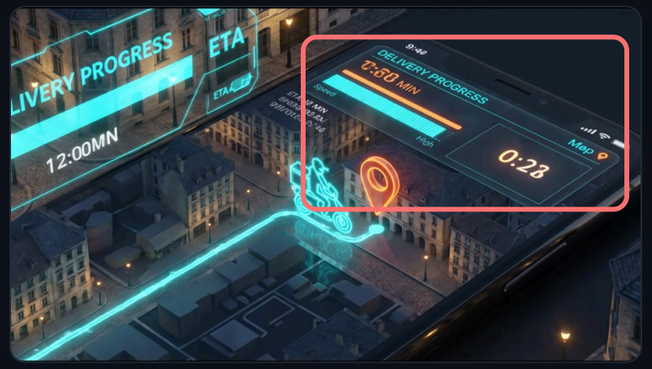
Kling (web): 3.0, 1080p with audio = 12 credits/s, i.e. 43,200 credits per hour; credit price depends on the plan. Chat apps: subscription, no per-second price.

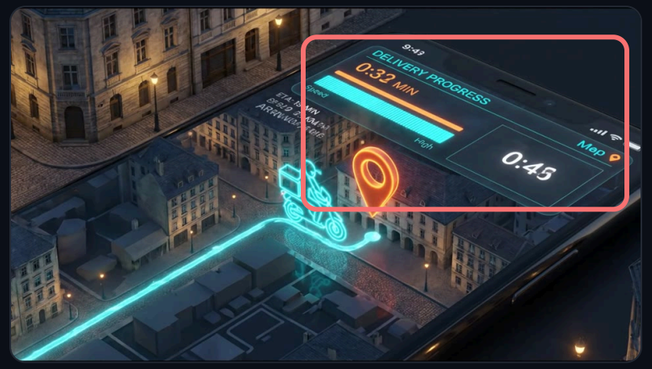
Prices: Gemini API, xAI API and Runway API, October 2026; Gemini Omni 1.1 Flash ≈$0.10/s (720p, token-billed), Grok Imagine Lite $0.02/s; only successful generations are billed. Not included: key frame images, editing, audio, people's time. 3 attempts per shot is an assumption for the estimate.

ACCEPTANCE

Acceptance checklist for a generated clip

Even a good image-to-video clip gets a frame-by-frame review. Defects from **our** shots 2 and 3:





Shot 3, 1.5 seconds apart: the timer jumps 0:28 → 0:45, “Mep” instead of “Map”, the icon is a moped while the courier rides a bicycle.



Shot 2: the “THABIK” sign is random letters.

ACCEPTANCE CHECKLIST

1. **Hands and objects** in contact: no merging or vanishing
2. **Text and numbers**: readable and stable
3. **Face and clothes**: match the reference
4. **Motion**: speed and direction (wheels forward)
5. **First frame**: matches the approved one
6. **Shot joins**: light and colour match

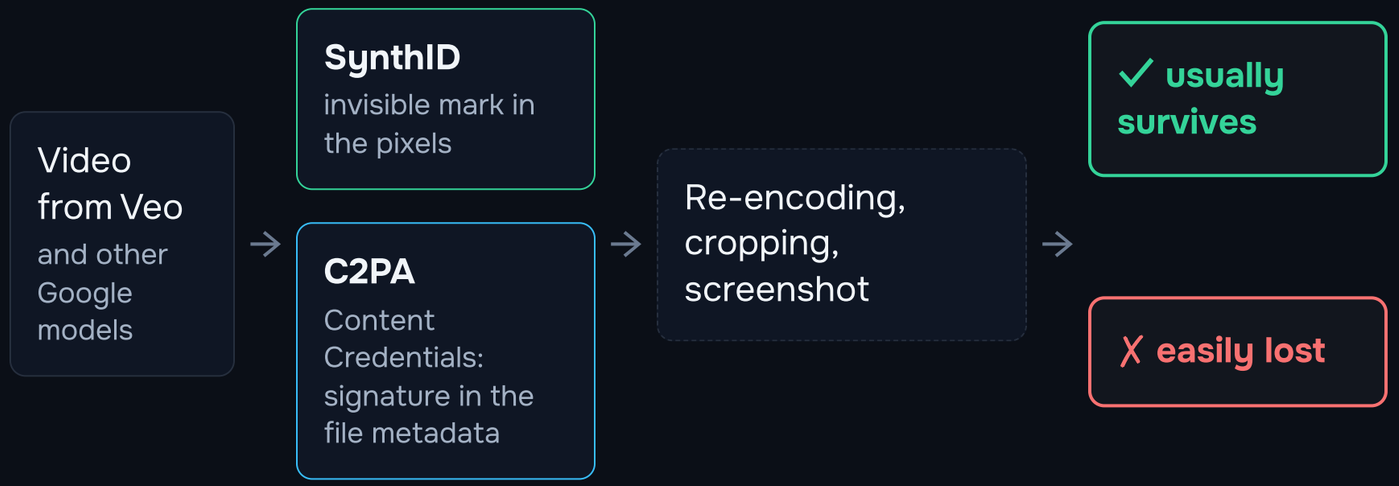
> The same checklist is the **homework rubric**.

RISKS

Rights, labelling and origin marks

- **Your own frames and characters**
  allowed, with an “AI-generated” label
- **Other brands, logos, styles**
  only with permission and after review by company lawyers
- **Real people without consent**
  employees, clients, public figures: only with written consent

HOW A MARK SURVIVES RE-SAVING



SynthID check: upload the video to the Gemini app (up to 100 MB, 90 s). It detects SynthID only; videos from other services may not be recognised.

EU AI Act, Art. 50: from 2 August 2026, whoever publishes a deepfake must label it, and the model provider must add a machine-readable mark. Fines up to €15M or 3% of turnover.

PRACTICE

Homework 5: a three-shot mini story



Track A · product

3 promo shots from your L04 frames: reveal, detail, final shot. Shape and text stay stable.



Track B · courier

3 shots of the courier story: pickup, ride, delivery. Face and uniform match the reference.

1. **Storyboard**, 3 rows → `storyboard.md`
2. **Text-to-video** with one prompt, as a baseline
3. 3 **image-to-video** shots; one of them also in a **second service**
4. Edit into `final.mp4`, 15–25 s
5. Prompts, settings, cost → `video_spec.md`; review with slide 18 → `video_qa.md`

```
genai-homeworks/L05/
├── storyboard.md
├── videos/  # shots + final.mp4
├── video_spec.md
└── video_qa.md
```

Reviewer: **ihar_rubanovich@epam.com**. Free allowances are small and change often: check before you start (table in the homework file).

SUMMARY

Lesson summary

1. **Storyboard first**
   the story is split into shots before any generation
2. **Approved key frame before video generation**
   image-to-video from an approved frame
3. **One action per shot**
   4–8 seconds; the story comes from the storyboard and editing
4. **Structured prompt**
   camera + action + motion details
5. **Acceptance and budget**
   a checklist for every clip; cost is per second



Lesson 06: sound and speech. Current video models generate audio themselves. Controlling it is covered in L06.